LIMITS QUIZ

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Name: Solutions

Use the function $f(x) = \dfrac{x^2 - 9}{x^2 + 2x - 3}$ to answer the following questions.

1. Use the Limit Laws to compute $\lim_{x \to -3} f(x)$.

$$\lim_{x \to -3} \frac{x^2-9}{x^2+2x-3} = \lim_{x \to -3} \frac{(x+3)(x-3)}{(x+3)(x-1)} = \lim_{x \to -3} \frac{x-3}{x-1} = \frac{-3-3}{-3-1} = \frac{-6}{-4} = \boxed{\frac{3}{2}}$$

2. Use the Limit Laws to compute $\lim_{x \to 1} f(x)$.

This limit does not exist:

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \frac{x-3}{x-1} = -\infty$ because $x-3<0$ near 1 and $x-1>0$.

$\lim_{x \to 1^-} f(x) = +\infty$ because $x-3<0$ near 1 and $0<x-1$.

Date: February 8, 2019.